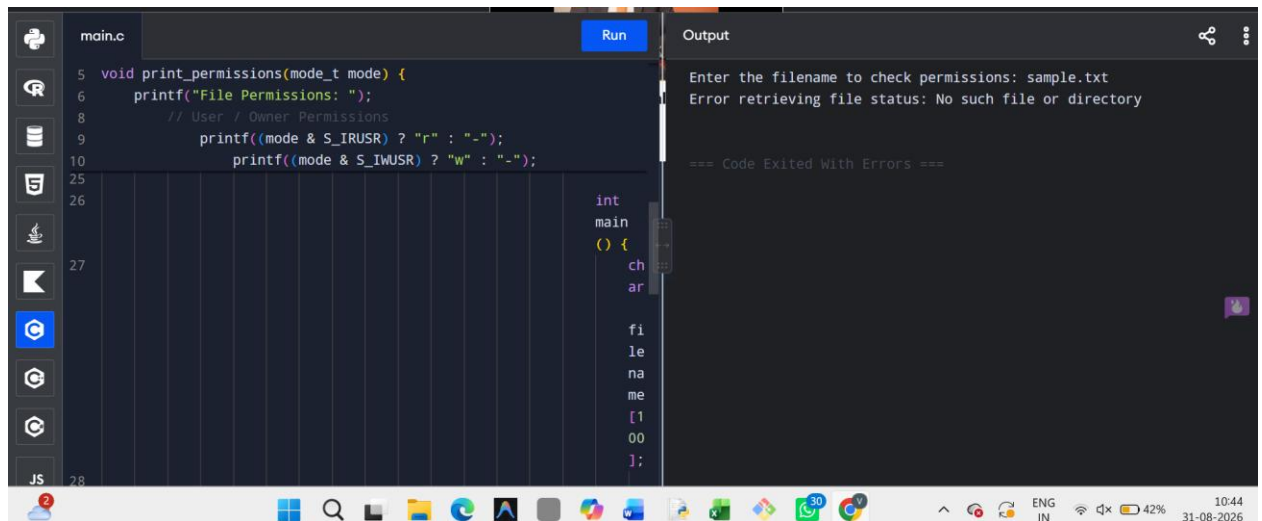


EXPERIMENT 34:

34. Illustrate the various File Access Permissions and different types of users in Linux.



```
main.c
5 void print_permissions(mode_t mode) {
6     printf("File Permissions: ");
7     // User / Owner Permissions
8     printf((mode & S_IRUSR) ? "r" : "-");
9     printf((mode & S_IWUSR) ? "w" : "-");
10
25
26
27
28
int
main
() {
    char
    filename[100];
}
```

```
Output
Enter the filename to check permissions: sample.txt
Error retrieving file status: No such file or directory

=== Code Exited With Errors ===
```

PSEUDO CODE :

- Define a target file path to inspect (passed via command line or user input).
- Call the `stat()` system call to populate a `stat` structure with file metadata.
- Check the `st_mode` field using bitwise `&` operations against standard permission macros (`S_IRUSR`, `S_IWUSR`, `S_IXUSR`, `S_IRGRP`, etc.).
- Construct and display a permission string representing the rights for the **Owner** (User), **Group**, and **Others**.
- Display the UID (User ID) and GID (Group ID) associated with the file owner.

CODE:

```
#include <stdio.h>
```

```
#include <sys/stat.h>
```

```
#include <unistd.h>
```

```
void
```

```
print_permissions(mode
```

```
_t mode) {
```

```
    printf("File  
Permissions: ");
```

```
    // User / Owner
```

```
Permissions
```

```
    printf((mode &  
S_IRUSR) ? "r" : "-");
```

```
    printf((mode &  
S_IWUSR) ? "w" : "-");
```

```
    printf((mode &  
S_IXUSR) ? "x" : "-");
```

```
    // Group Permissions
```

```
    printf((mode &  
S_IRGRP) ? "r" : "-");
```

```
    printf((mode &  
S_IWGRP) ? "w" : "-");
```

```
    printf((mode &  
S_IXGRP) ? "x" : "-");
```

```
    // Others Permissions
```

```
    printf((mode &  
S_IROTH) ? "r" : "-");
```

```
    printf((mode &  
S_IWOTH) ? "w" : "-");
```

```
    printf((mode &  
S_IXOTH) ? "x" : "-");
```

```
    printf("\n");  
}
```

```
int main() {
```

```
    char filename[100];
```

```
    struct stat file_stat;
```

```
printf("Enter the  
filename to check  
permissions: ");
```

```
scanf("%s",  
filename);
```

```
// Fetch file status
```

```
if (stat(filename,  
&file_stat) < 0) {
```

```
    perror("Error  
retrieving file status");
```

```
    return 1;
```

```
}
```

```
printf("\n--- File  
Information ---\n");
```

```
printf("File Path:  
%s\n", filename);
```

```
printf("Owner UID:  
%d\n", file_stat.st_uid);
```

```
printf("Group GID:  
%d\n", file_stat.st_gid);
```

```
print_permissions(file_s  
tat.st_mode);
```

```
return 0;
```

}INPUT :

Enter the filename to

check permissions:

sample.txt

OUTPUT:

--- File Information ---

File Path: sample.txt

Owner UID: 1000

Group GID: 1000

File Permissions: rw-r--

r--